

Anesa Ibrahimi

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GitHub Profile
LinkedIn Profile

SUMMARY

Artificial Intelligence graduate with published research in vision-language models and hands-on experience developing deep learning systems across computer vision, multimodal AI, and information retrieval. Experienced in transformer-based architectures, large-scale training on HPC infrastructure, experimental evaluation of machine learning systems, and robotics perception systems. Seeking roles in Machine Learning, AI Engineering, or Applied AI.

TECHNICAL SKILLS

Languages: Python, SQL, MATLAB
Machine Learning / Deep Learning: PyTorch, TensorFlow, HuggingFace Transformers, Keras
Tools & Infrastructure: Docker, Git, CUDA, Slurm (HPC), Linux, Jupyter, 3D Slicer
Specializations: Computer Vision, Multimodal AI, Information Retrieval, Reinforcement Learning

EDUCATION

- Universiteit van Amsterdam** 2023-2025
MSc. Artificial Intelligence Amsterdam, Netherlands
- Vrije Universiteit Amsterdam** 2020-2023
BSc. Artificial Intelligence, Specialisation : Intelligent Systems, Minor: Data Science Amsterdam, Netherlands

PROFESSIONAL EXPERIENCE

Actively seeking Machine Learning Engineering / AI Engineering roles

TECHNICAL EXPERIENCE

- Vrije Universiteit Amsterdam** March 2023 - May 2023
Professor's Assistant - Databases Amsterdam
Professor: Dr. J. Endrullis
 - Delivered weekly database labs to ~ 50 undergraduate students, covering relational modeling and SQL query optimization.
 - Mentored students on relational schema design and SQL queries, supporting project completion across multiple assignments.
- Vrije Universiteit Amsterdam** March 2022 - May 2022
Professor's Assistant - Human-Computer Interaction Amsterdam
Professor: Dr. E.M.A.L. Beauxis Aussalet
 - Led weekly HCI practical sessions for students covering interactive system design and prototyping workflows.
 - Guided teams in applying usability testing and evaluation methods for interactive systems.
 - Provided technical feedback on project prototypes and UX evaluation reports.

TECHNICAL PROJECTS

- Fine-grained image understanding with Vision-Language Models** January 2025 - August 2025
Master Thesis Project
 - Designed and implemented a novel **Memory-Augmented Vision Encoder** by integrating trainable key-value memory modules into the MLP layers of a Vision Transformer (ViT).
 - Developed a two-stage training pipeline leveraging **knowledge distillation** to transfer representations from a pre-trained CLIP (teacher) to the memory-enhanced student model, enabling efficient training.
 - Achieved state-of-the-art performance, decisively outperforming the baseline CLIP model on **long-caption, fine-grained retrieval benchmarks**.
 - Trained and evaluated on HPC infrastructure across **Snellius** (4x NVIDIA A100/H100) and **DAS-5** partitions using Slurm for multi-GPU data parallel experiments.
 - Technologies:** Python, PyTorch, CUDA, OpenAI CLIP, Vision Transformers, Slurm/HPC (Snellius, DAS-5).
- Multi-Modal NAO Robot** October 2024 - December 2024
 - Led a 7-member team in developing a **multi-modal emotion recognition and response system** for the NAO robot.
 - Engineered a distributed multimodal perception pipeline combining RoBERTa (text), SpeechBrain (audio), and LLaMA Vision models with OpenVoice emotional speech synthesis.

- Conducted controlled human–robot interaction experiments with **10 participants and 20 interaction trials**, achieving **60%** exact emotion classification accuracy and **90%** accuracy at emotional category level.
 - **Technologies:** Python, PyTorch, Docker, OpenVoice, HuggingFace models, NAO Framework.
- **LLM4CS** *October 2024 - December 2024*
 - Reproduced and extended the LLM4CS framework to evaluate large language models as **context-aware search intent interpreters** in multi-turn conversational search tasks.
 - Benchmarked prompting strategies and response aggregation methods across **dense (Faiss) and sparse (BM25) retrieval pipelines** using Pyserini.
 - Analyzed retrieval effectiveness of LLM-assisted conversational search pipelines across multiple query reformulation strategies.
 - **Technologies:** Python, Pyserini, Information Retrieval, LLMs, Prompt Engineering, Dense & Sparse Retrieval.
 - **AI for Medical Imaging (AI4MI) – SegTHOR Challenge** *September 2024 - October 2024*
 - Developed CNN-based models for segmenting **thoracic organs-at-risk** (esophagus, heart, trachea, aorta) from 3D CT scans in the SegTHOR dataset (60 patients).
 - Improved ENet segmentation performance through loss-function optimization, data augmentation, and regularization tuning.
 - Benchmarked multiple architectures (ENet, UNet, UNet++, DeepLabV3+) using **Dice, Jaccard, Hausdorff Distance, Precision/Recall**.
 - **Technologies:** Python, PyTorch, Medical Image Segmentation, 3D Slicer, CNN Architectures.
 - **In-Context Learning Improves Compositional Understanding of Vision-Language Models** *April 2024 - May 2024*
Published at ICML 2024 Workshop on Foundation Models in the Wild
 - Conducted a systematic benchmark of contrastive (CLIP-based) and generative VLMs across compositional reasoning datasets including Winoground, ARO, and SugarCrepe.
 - Designed an in-context learning (ICL) prompting pipeline using synthetic and COCO images, improving compositional reasoning performance by up to **~6% on ARO benchmarks**.
 - First-authored a **peer-reviewed workshop paper at ICML 2024**, presenting insights into architectural limitations and prompting strategies for vision-language compositionality.
 - **Technologies:** Python, PyTorch, Vision-Language Models, CLIP, LLaVA, CogVLM, In-Context Learning
 - **Instance Diffusion - Extension** *April 2024 - May 2024*
 - Reproduced the InstanceDiffusion framework for instance-controlled image generation using the MSCOCO dataset and diffusion-based models.
 - Developed an LLM-powered module to automatically generate image captions and bounding boxes, eliminating manual annotation in the generation pipeline.
 - Improved bounding-box detection performance in reproduction experiments.
 - **Technologies:** Python, PyTorch, Diffusion Models, Stable Diffusion, CLIP, MSCOCO
 - **Optimizing Locomotion with Evolutionary Algorithms in MuJoCo** *March 2023 - June 2023*
Bachelor Thesis Project – Vrije Universiteit Amsterdam
 - Explored the potential of Evolutionary Algorithms (CMA-ES, XNES, SNES) for solving continuous control optimization problems in the MuJoCo Ant environment.
 - Designed and implemented a full simulation and optimization pipeline for evolving the ant agent’s locomotion behavior across 100 virtual environments with varying terrain densities and obstacle heights.
 - Conducted quantitative and qualitative analyses to benchmark the performance of each algorithm.
 - **Technologies:** Python, MuJoCo, OpenAI Gym, EvoTorch, NumPy, Pandas, Matplotlib.

PUBLICATIONS

- **Anesa Ibrahim**, Matteo Nulli, Avik Pal, Hoshe Lee, Ivona Najdenkoska, *In-Context Learning Improves Compositional Understanding of Vision-Language Models*, arXiv, 2024. <https://arxiv.org/abs/2407.15487>
 - Paper published at ICML Foundation Models in the Wild Workshop